JONGIN CHOI

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EDUCATION

Pennsylvania State University ***May 2025***

Bachelor of Science, Applied Data Sciences *University Park, PA*

- Relevant Coursework: Data Analysis, Software Engineering, Statistics, Algorithms, Artificial Intelligence, Machine Learning

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript

Data & ML: Pandas, NumPy, Scikit-learn, PyTorch, HuggingFace Transformers, Jupyter, Spark

Web & Scraping: Beautiful Soup, Selenium, HTML/CSS

Databases: MySQL, MongoDB

Tools: Git, Linux, Docker

EXPERIENCE

Research Assistant — Multimodal Emotion Dataset (FeelReel) ***Jun 2024 – Mar 2025***

Pennsylvania State University, Data Science Lab (Prof. Sharon Hwang) · IST 495 internship credit *University Park, PA*

- Supported the follow-up phase of FeelReel, a lab benchmark of 400K+ Reddit images and 5M+ comments for emotion understanding (earlier version submitted to CVPR 2024).
- Sourced and organized Reddit image posts and comments through the project's data collection pipeline, keeping datasets structured and reproducible.
- Built the MTurk survey (HIT) page and collected human evaluation responses on the dataset's emotion annotations through it.

PROJECTS

FeelReel-Mini — Personal recreation ***Jul – Oct 2026***

github.com/nopgae/feelreel-mini *PyTorch, Hugging Face, PRAW, LLaVA (Ollama), matplotlib*

- Rebuilt the FeelReel data pipeline at small scale with public tools: Reddit image posts and comments, GoEmotions pseudo-labels (RoBERTa, 28 emotions), and LLaVA 7B asked what in each image triggered the emotion.
- Charted emotion distributions across 8 image-heavy subreddits: scenery and woodworking communities leaned toward admiration and surprise, pet communities toward joy.
- Separate from the lab's code and data, which are not public.

Medical Transcription Classifier — Penn State DS 310 ***Fall 2024***

- Classified medical transcriptions into 6 specialties using TF-IDF (1–3 grams) plus document-length features; final weighted F1 0.807.
- Tuned XGBoost over 54 configurations with 5-fold CV (270 fits), then showed untuned logistic regression still won (F1 0.763 vs 0.711) and traced it to sparse features.

Reddit Sentiment Pipeline — Personal

github.com/nopgae/reddit-sentiment-roberta *PyTorch, Hugging Face, pandas, Plotly*

- Built an end-to-end pipeline alone: Reddit JSON API collection, batched inference with a pre-trained RoBERTa sentiment model, aggregation and charts.
- Scored 563 posts across 7 e-commerce, logistics and supply-chain subreddits; quantified community baselines (r/supplychain +0.11 vs r/ecommerce –0.31).

Crime & Weather Data Analysis — Chicago ***Sep 2024***

- Analyzed correlations between crime rates and weather conditions by integrating City of Chicago crime records with daily weather data.
- Used SQL to merge heterogeneous datasets and performed data cleaning to ensure analytical consistency.

ADDITIONAL INFORMATION

Languages: Fluent in English and Korean (bilingual); TOEIC Speaking 160 (AL)

Interests: Web security CTF (webhacking.kr), Boston Dynamics Spot robotics, Korean stock market analysis